MADHEPURA ELECTRIC LOCOMOTIVE PROJECT (WAG_{12B})

Adhesion Calculations

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0 DOCUMENT INFORMATION

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5	Benu	02/05/2023	8,8.3,8.4 , 8.7 & 8.8	Updated Haulage chart for Wet track with improved adhesion	N/A

0.2 Confidentiality clause

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1 INTRODUCTION

1.1 Purpose and Scope

Purpose and scope of this document is to provide the adhesion calculations of WAG_{12B} as per the Schedule G Annexure I of the Agreement for PCMA of Madhepura Electric Locomotive dated January 2014.

1.2 Traceability Matrix

Table below shows the technical traceability of the technical requirement related to the Traction Subsystem with reference to the Schedule A Specifications and Standards.

Schedule A – Standards & Specification Section Number	Alstom Requirement ID	Description of Requirement	Reference Section in this document	Remarks
Clause 2.4 of performance requirements	CTS-4220	Under dry rail conditions, the Locomotive shall be able to generate tractive effort during start and at low speeds corresponding to at least 40% adhesion	Sec 3	
Clause 2.2 of performance requirements	CTS-4080	Pneumatic brake effort: 7% - 9% of gross weight	Sec5	

1.3 Abbreviations

Acronym / Abbreviation	Definition
E block1	Traction Converter with Integrated Auxiliary Inverter
E block2	Traction Converter
Locomotive	BoBo + BoBo
Section	Half locomotive (one BoBo)

2 GENERAL DESCRIPTION

Below Fig shows WAG_{12B} locomotive in Twin BOBO Configuration.

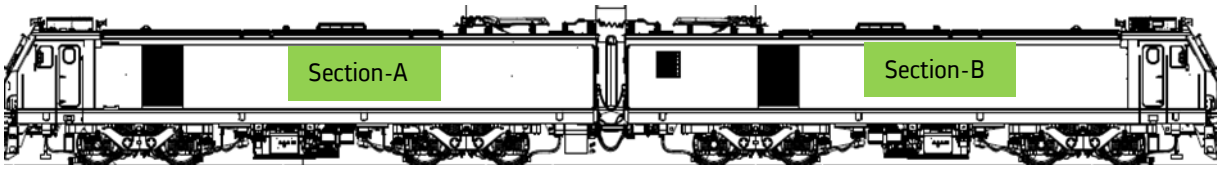


Figure 1: Twin BOBO Configuration of WAG_{12B} Locomotive

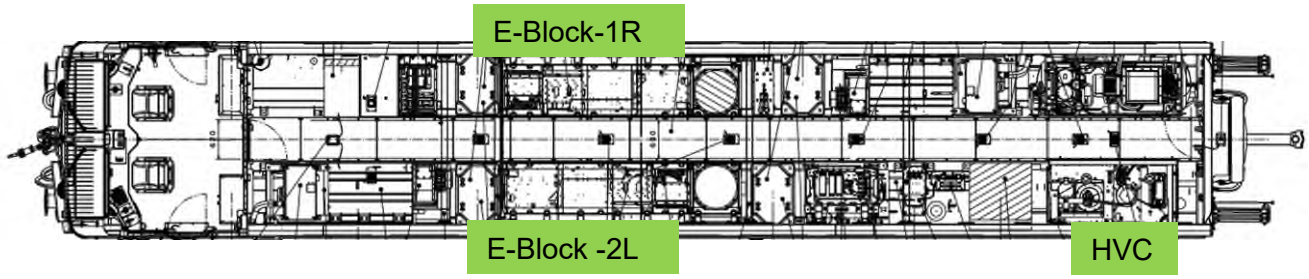


Figure 2: Machine Room - One Section of WAG_{12B} Locomotive

Updated the figure

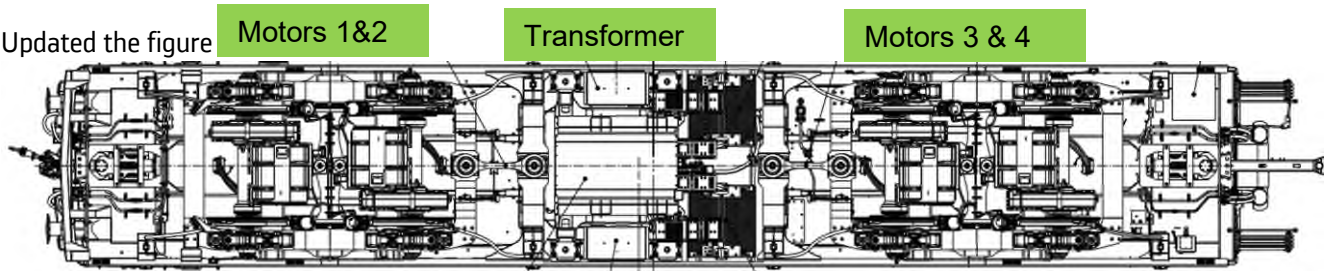


Figure 3: Bottom view - One Section of WAG_{12B} Locomotive

3 ADHESION PERFORMANCES DURING TRACTION

3.1 Traction Performance under 25 T/Axle

Figure 4 shows the traction performances with 25 T/axle load. Traction performances are shown under normal mode and degraded mode along with resistance to motion at different slopes (flat track, 1/150, 1/250 and 1/500).

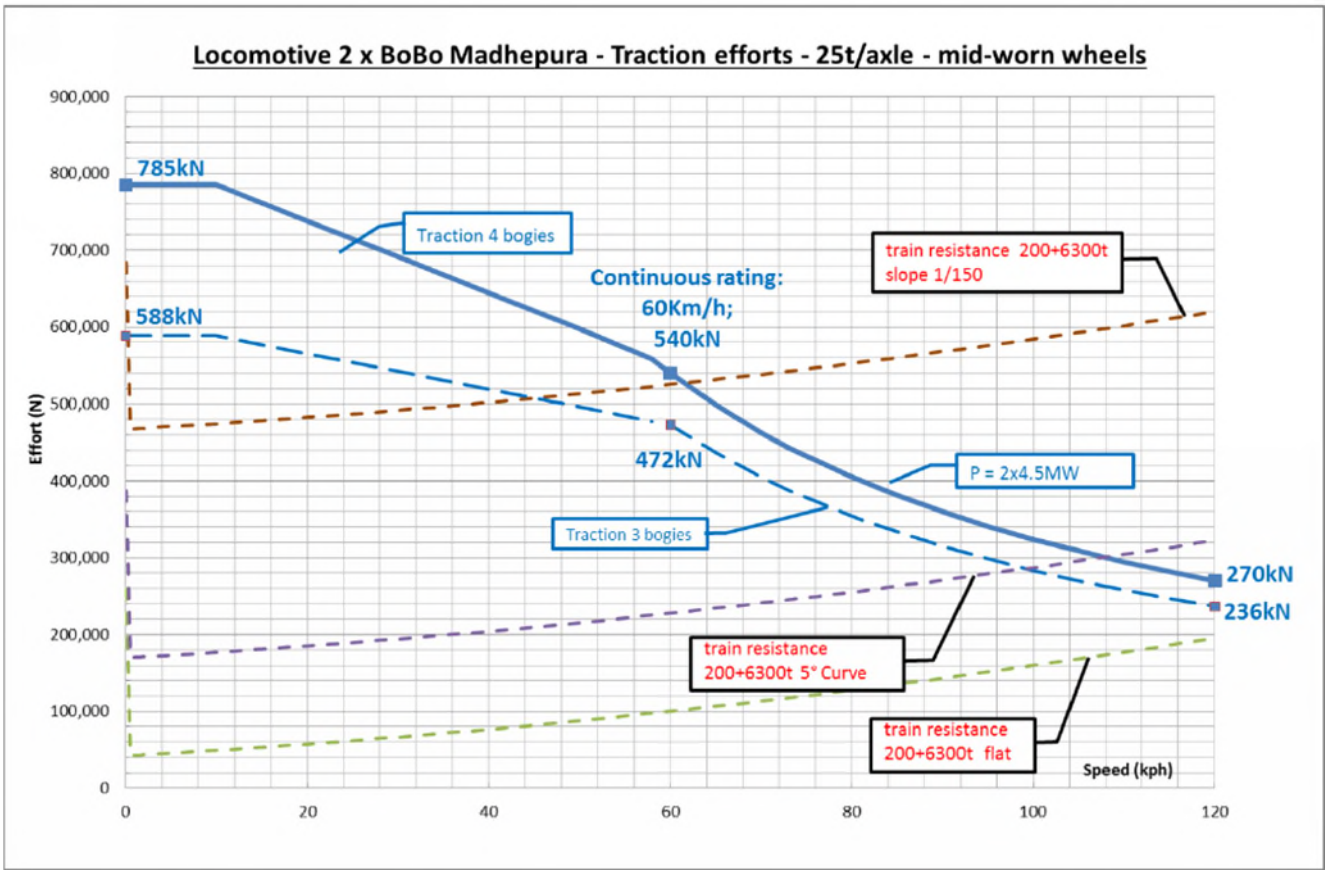


Figure 4: Traction Performance under 25 T/Axle

Adhesion at starting with 4 Bogies:

$$\%Adhesion = \frac{Effort}{9.81 \ * \ Mass\ of\ the\ locomotive} * 100 = \frac{785\ kN}{9.81 * 25 * 8} * 100 = 40\%$$

Adhesion at starting with 3 Bogies:

$$\%Adhesion = \frac{Effort}{9.81 \ * \ Mass\ of\ the\ locomotive} * 100 = \frac{588\ kN}{9.81 * 25 * 6} * 100 = 40\%$$

3.2 Adhesion Performance under 25 T/Axle

Figure 5 shows the adhesion performance with respect to speed during normal mode and degraded mode with 25 T/axle load.

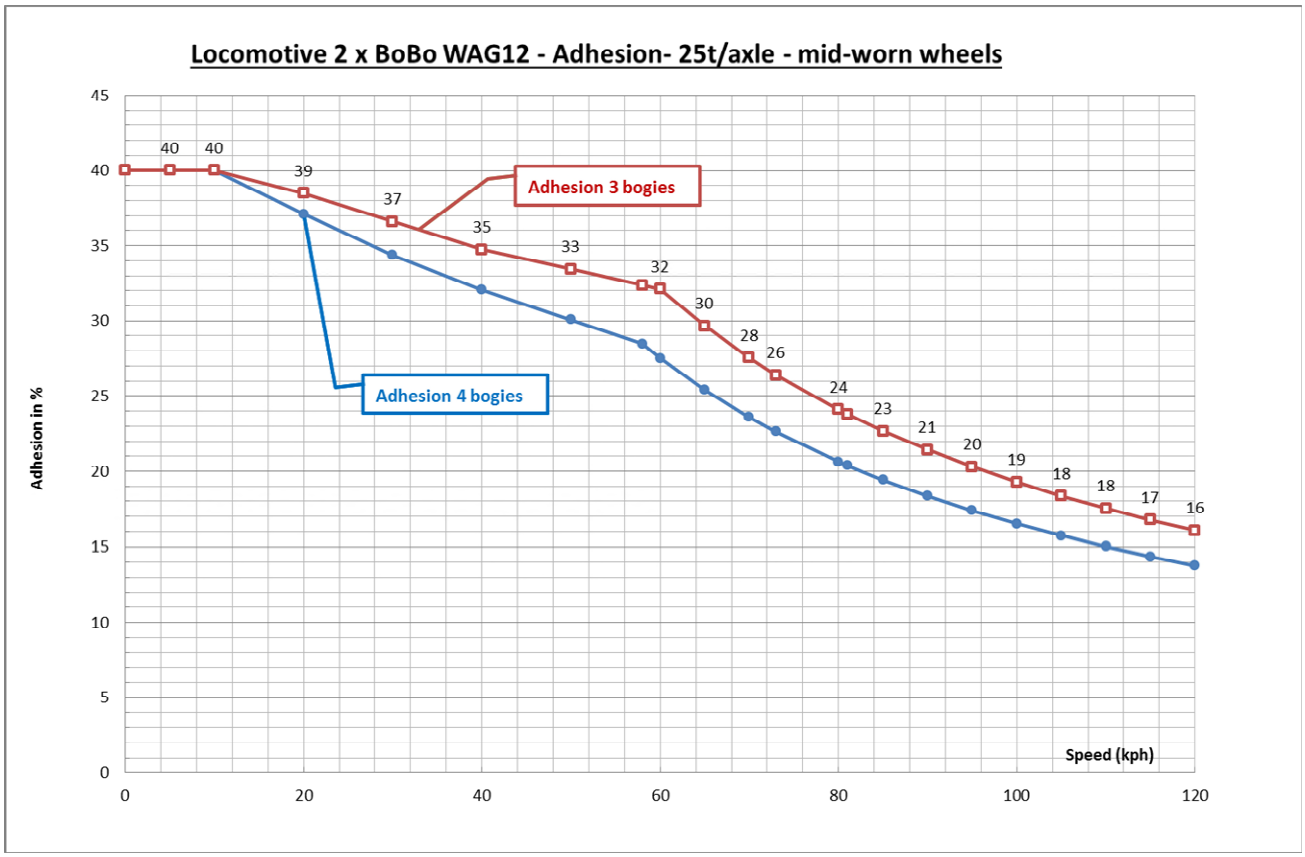


Figure 5: Adhesion Performance under 25 T/Axle

From the figure 5, it is evident that the starting adhesion is 40% under normal and degraded modes of operation.

3.3 Traction Performance under 22.5 T/Axle

Figure 6 shows the traction performances with 22.5 T/axle load. Traction performances are shown under normal mode and degraded mode along with resistance to motion at different slopes (flat track, 1/150, 1/250 and 1/500).

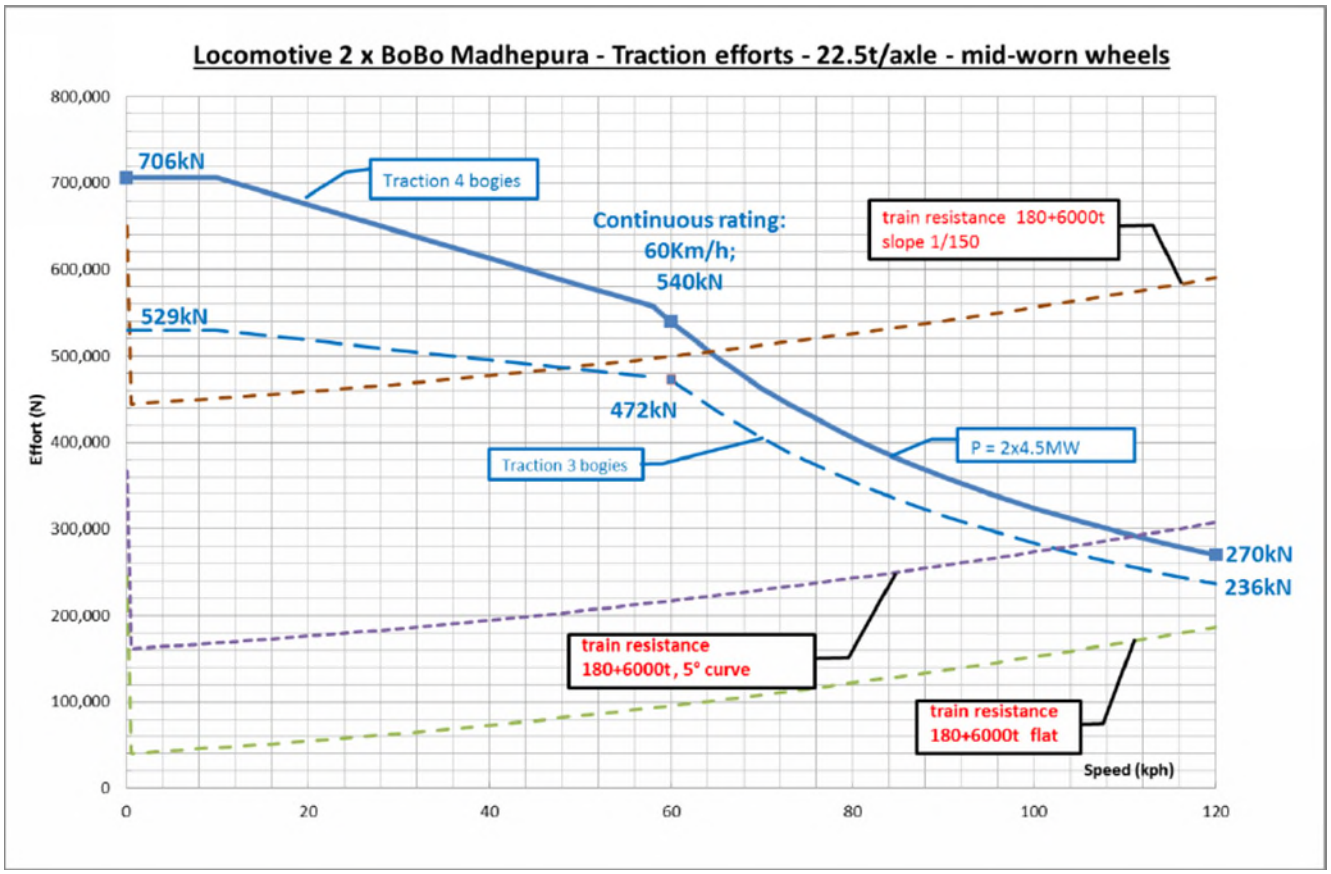


Figure 6: Traction Performance under 22.5 T/Axle

Adhesion at starting with 4 Bogies:

$$\%Adhesion = \frac{Effort}{9.81 \ * \ Mass\ of\ the\ locomotive} * 100 = \frac{706\ kN}{9.81 * 22.5 * 8} * 100 = 40\%$$

Adhesion at starting with 3 Bogies:

$$\%Adhesion = \frac{Effort}{9.81 \ * \ Mass\ of\ the\ locomotive} * 100 = \frac{529\ kN}{9.81 * 22.5 * 6} * 100 = 40\%$$

3.4 Adhesion Performance under 22.5 T/Axle

Figure 7 shows the adhesion performance with respect to speed during normal mode and degraded mode with 22.5 T/axle load.

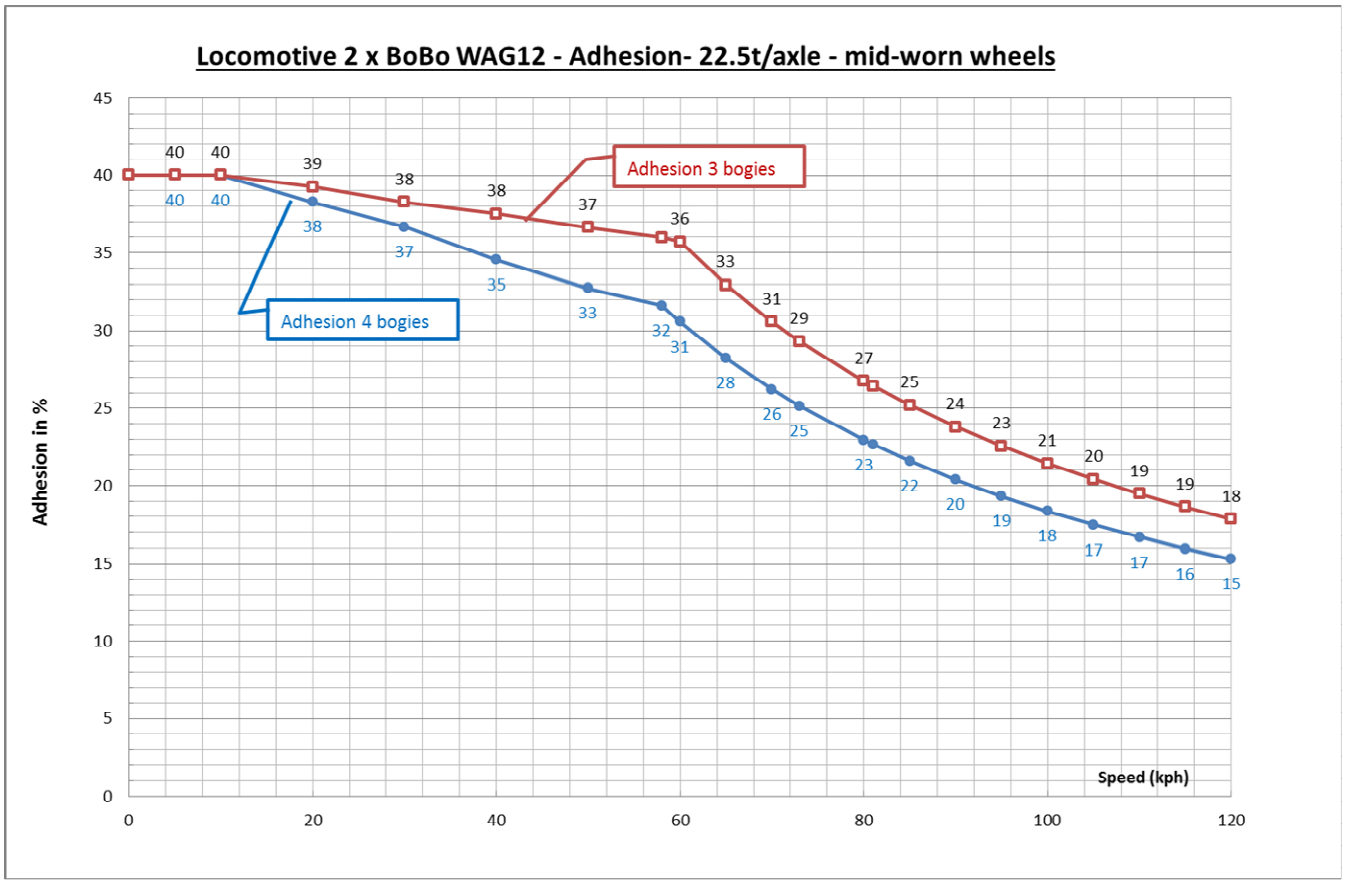


Figure 7: Adhesion Performance under 22.5 T/Axle

From the figure 7, it is evident that the starting adhesion is 40% under normal and degraded modes of operation.

4 ADHESION PERFORMANCES DURING ELECTRICAL BRAKING

4.1 Adhesion Performance under 25 T/Axle during Normal Mode

Figure 8 shows the electric braking performance with 25 T/axle load. Electric braking and adhesion performances are shown under normal mode.



Figure 8: Electric Braking & Adhesion Performance in normal mode with 25 T/Axle

Maximum Adhesion with 4 Bogies:

$$\%Adhesion = \frac{Effort}{9.81 \ * \ Mass\ of\ the\ locomotive} * 100 = \frac{563\ kN}{9.81 * 25 * 8} * 100 = 29\%$$

4.2 Adhesion Performance under 25 T/Axle during Degraded Mode

Figure 9 shows the electric braking performance with 25 T/axle load. Electric braking and adhesion performances are shown under degraded mode.

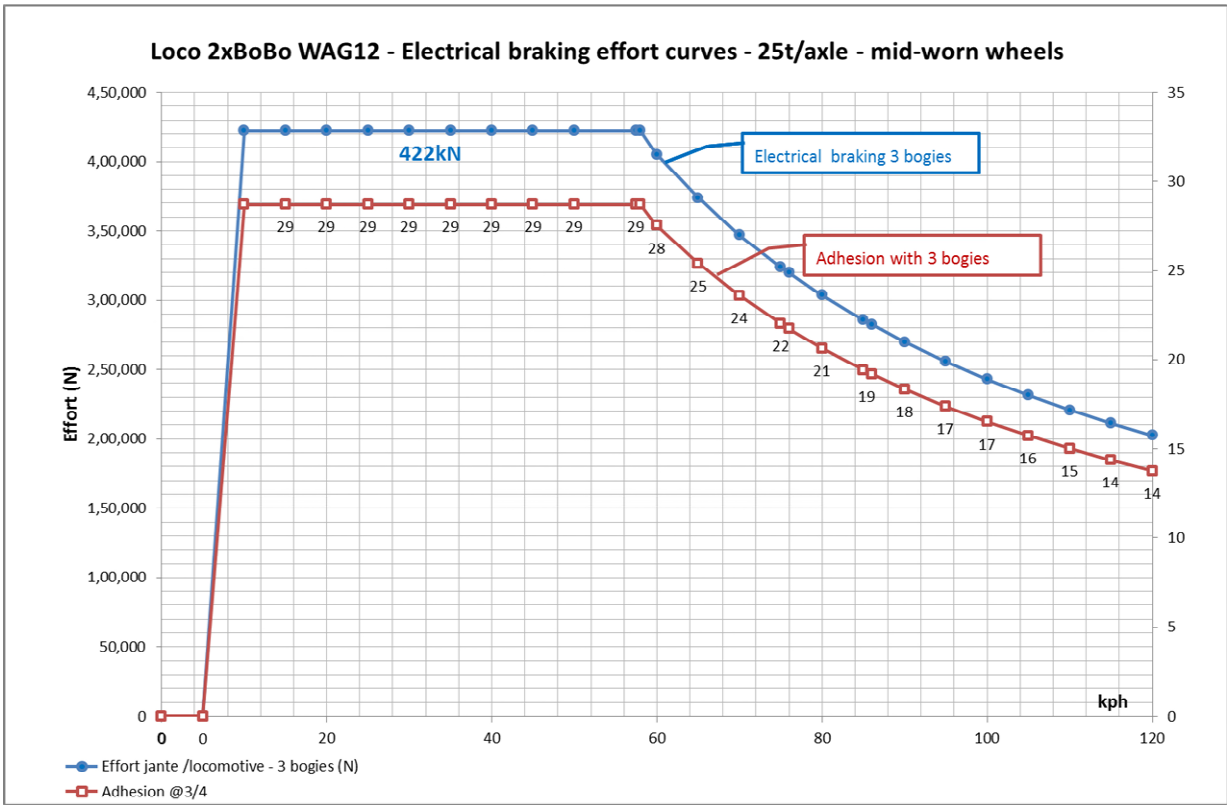


Figure 9: Electric Braking & Adhesion Performance in degraded mode with 25 T/Axle

Maximum Adhesion with 3 Bogies:

$$\%Adhesion = \frac{Effort}{9.81 \ * \ Mass\ of\ the\ locomotive} * 100 = \frac{422\ kN}{9.81 * 25 * 6} * 100 = 29\%$$

4.3 Adhesion Performance under 22.5 T/Axle during Normal Mode

Figure 10 shows the electric braking performance with 22.5 T/axle load. Electric braking and adhesion performances are shown under normal mode.

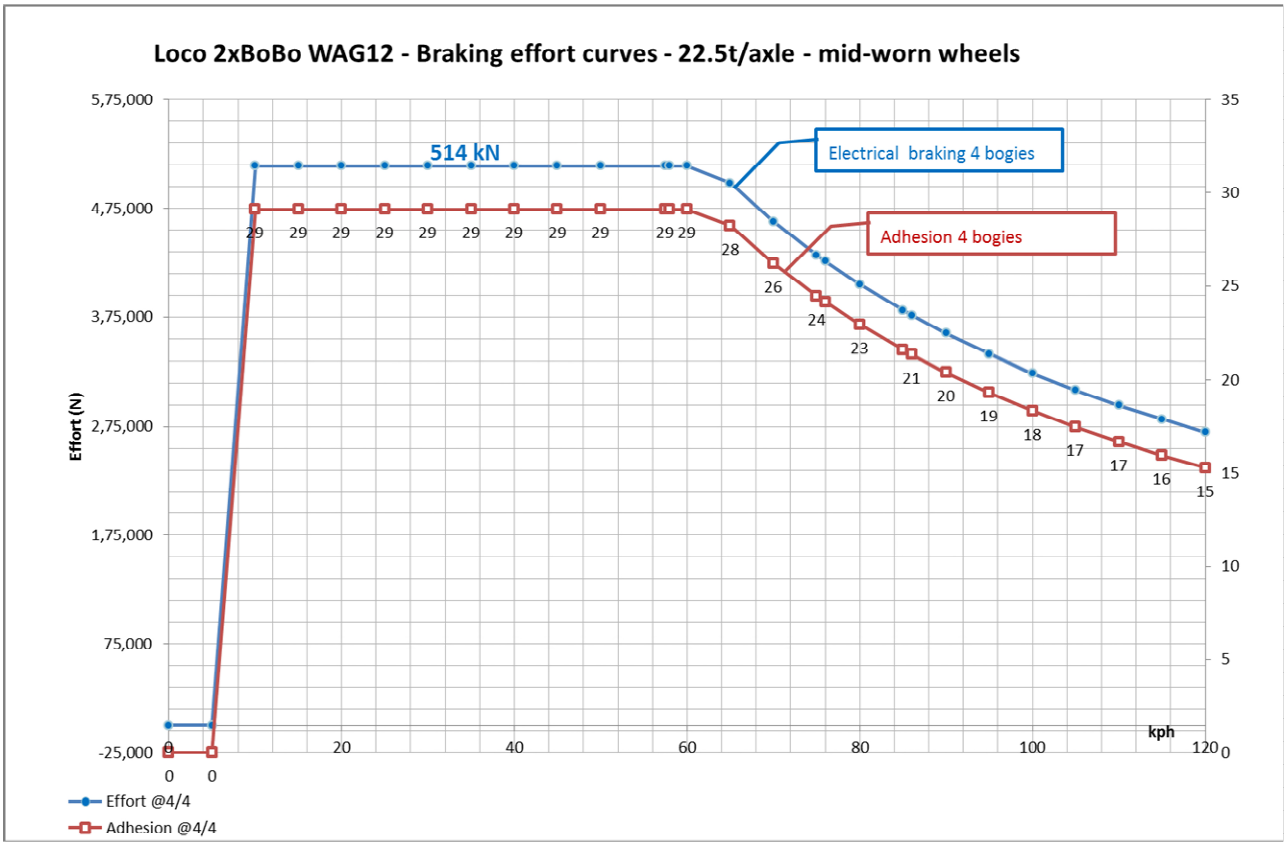


Figure 10: Electric Braking & Adhesion Performance in normal mode with 22.5 T/Axle

Maximum Adhesion with 4 Bogies:

$$\%Adhesion = \frac{Effort}{9.81 \ * \ Mass\ of\ the\ locomotive} * 100 = \frac{514\ kN}{9.81 * 22.5 * 8} * 100 = 29\%$$

4.4 Adhesion Performance under 22.5 T/Axle during Degraded Mode

Figure 11 shows the electric braking performance with 22.5 T/axle load. Electric braking and adhesion performances are shown under degraded mode.

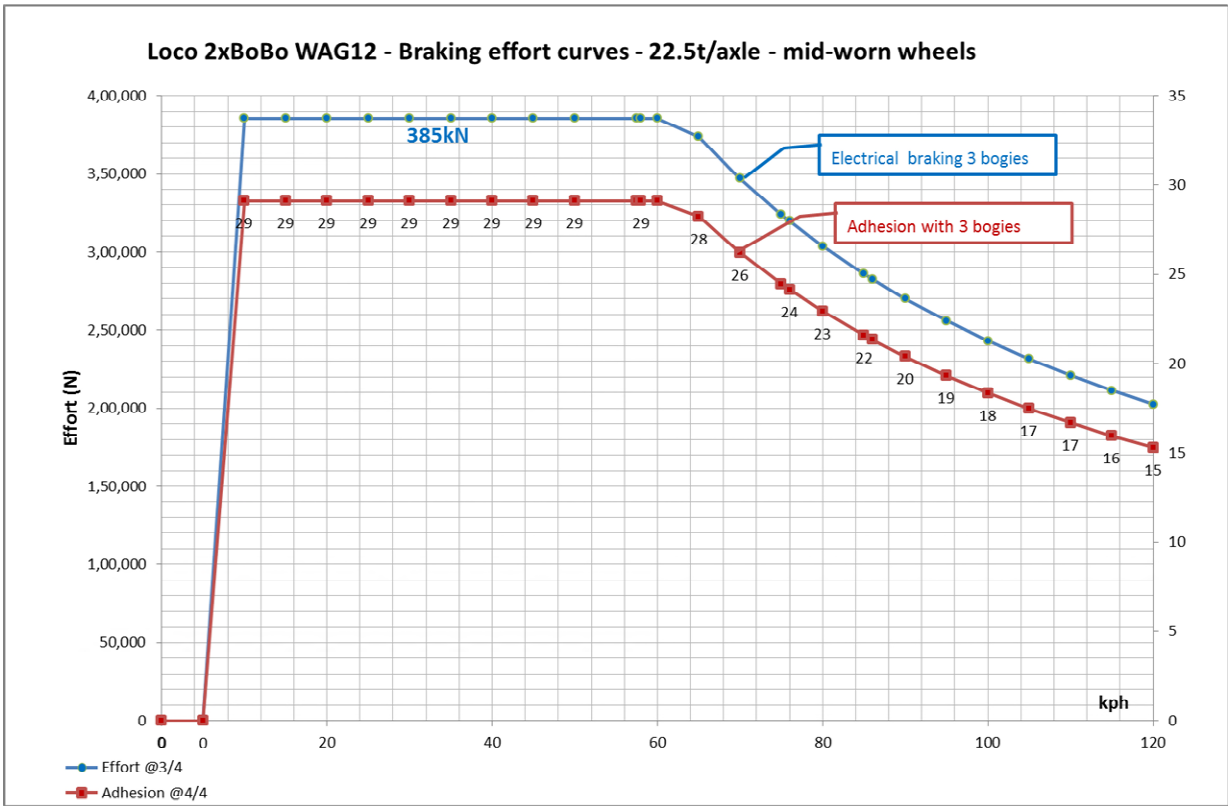


Figure 11: Electric Braking & Adhesion Performance in degraded mode with 22.5 T/Axle

Maximum Adhesion with 3 Bogies:

$$\%Adhesion = \frac{Effort}{9.81 \ * \ Mass\ of\ the\ locomotive} * 100 = \frac{385\ kN}{9.81 * 22.5 * 6} * 100 = 29\%$$

5 ADHESION UNDER TORQUE LIMITATIONS & FRICTION BRAKE:

Adhesion during two starting torque limitations is mentioned in the below table

S.No	Tractive Effort	Axle Weight	Adhesion
1	529kN	25 T	**27%**
2	529kN	22.5 T	**30%**
3	300kN	25 T	**15.3%**
4	300kN	22.5 T	**17%**

Table 1: Adhesion under Torque Limitations

Adhesion during friction brake is mentioned in the below table under different axle loads

S.No	Friction Brake Effort	Axle Weight	Adhesion
1	154kN	25 T	**7.8%**
2	154kN	22.5 T	**8.7%**

Table 2: Adhesion under Friction Brake

Adhesion calculation during friction brake:

- 25 t/gauge

$$\%Adhesion = \frac{Effort}{9.81 \ * \ Mass\ of\ the\ locomotive} * 100 = \frac{154\ kN}{9.81 * 25 * 8} * 100 = 7{,}8\%$$

- 22,5 t/gauge

$$\%Adhesion = \frac{Effort}{9.81 \ * \ Mass\ of\ the\ locomotive} * 100 = \frac{154\ kN}{9.81 * 22{,}5 * 8} * 100 = 8{,}7\%$$

6 MINIMUM EFFORT REQUIRED TO START THE LOCOMOTIVE

6.1 Starting Resistance:

a. 22.5 T/axle

- Weight of the locomotive = 180 T
- Starting resistance for locomotive on level track = 6 k.g/tonne
- Force required to start the locomotive on level track = 180 * 6 *9.81 = 10.595 kN

b. 25 T/axle

- Weight of the locomotive = 200 T
- Starting resistance for locomotive on level track = 6 k.g/tonne
- Force required to start the locomotive on level track = 200 * 6 *9.81 = 11.7 kN

6.2 Resistance to Motion on level track

V(Km/h)	Resistance to Motion in N @22.5T/axle	Resistance to Motion in N @25T/axle
0	10595	11772
2	2211	2342
4	2251	2385
6	2295	2433
8	2344	2485
10	2397	2542
12	2454	2603
14	2516	2669
16	2583	2739
18	2654	2814
20	2729	2893
22	2809	2976
24	2894	3064
26	2982	3157
28	3076	3254
30	3174	3355
32	3276	3461
34	3383	3572
36	3494	3687
38	3610	3806
40	3730	3930
42	3854	4058
44	3984	4191
46	4117	4328
48	4255	4470
50	4398	4616
52	4545	4767
54	4696	4922
56	4852	5082
58	5013	5246

V(Km/h)	Resistance to Motion in N @22.5T/axle	Resistance to Motion in N @25T/axle
60	5178	5414
62	5347	5587
64	5521	5765
66	5699	5947
68	5882	6133
70	6069	6324
72	6261	6520
74	6457	6720
76	6658	6924
78	6863	7133
80	7073	7346
82	7287	7564
84	7505	7786
86	7729	8013
88	7956	8244
90	8188	8480
92	8425	8720
94	8666	8965
96	8911	9214
98	9161	9467
100	9415	9725
102	9674	9988
104	9937	10255
106	10205	10526
108	10478	10802
110	10754	11083
112	11035	11367
114	11321	11657
116	11611	11951
118	11906	12249
120	12205	12552

Effort required to start the locomotive on level track for 22.5 T/axle and 25T/axle are provided above and the resistance to motion table provides minimum effort required to run the locomotive from the starting speed to 120kmph for both the axle weights (22.5T and 25T).

7 EFFORT-SPEED CURVE OF THE LOCOMOTIVE

Effort –Speed Curve of locomotive at full traction position from 0 to 120kmph

7.1 Normal Mode (8 Motors working):

Speed in Kmph	Effort in N 25T/axle	Effort in N 22.5T/axle
0	785000	706068
5	785000	706068
10	785000	706068
20	728217	675785
30	674690	647365
40	629014	610143
50	590017	578217
58	558621	558621
60	540000	540000
65	498462	498462
70	462857	462857
73	443836	443836
80	405000	405000
81	400000	400000
85	381176	381176
90	360000	360000
95	341053	341053
100	324000	324000
105	308571	308571
110	294545	294545
115	281739	281739
120	270000	270000

7.2 Degraded Mode (6 Motors working):

Speed in Kmph	Effort in N 25T/axle	Effort in N 22.5T/axle
0	588750	529500
5	588750	529500
10	588750	529500
20	565536	518136
30	542322	506772
40	519108	495408
50	495894	484044
58	477323	474954
60	472680	472680
65	436320	436318
70	405154	405154
73	388504	388504
80	354510	354510
81	350133	350136
85	333656	333656
90	315120	315120
95	298535	298534
100	283608	283608
105	270103	270102
110	257825	257826
115	246616	246614
120	236340	236340

8 HAULAGE CAPACITY OF THE LOCOMOTIVE

Below are the load haulage charts under dry and wet conditions with 22.5 T/axle and 25T/axle loads. Under dry condition adhesion considered is 40%. The load haulage results are indicative in nature and actual load haulage capability depends on the available adhesion between rail and wheel and actual load on the locomotive.

As per RDSO document MP – Guide 26 (Section 6.2), *"The adhesion will be lower by around 15% - 20% on wet rail compared to dry rails. Similarly, adhesion on oily rail will be much lower compared to dry rail as much 40%.* In view of the same this document considers 32% as nominal wet adhesions, 24% as and when operated through oily/ deteriorated wet conditions like dry leaf, insects, dust or combinations etc.. The below wet haulage charts are prepared based on 20% reduction from dry adhesion of 40% (32% adhesion).

The Locomotive operations also need to ensure sufficient sand availability, plan for bankers based on the actual weather conditions existing. Additionally appropriate attacking speeds are used and included in the working time table (WTT). Also appropriate driving behaviors while commuting through gradients & difficult weather conditions of Loco Pilot to be part of training.

8.1 Axle Weight 22.5 T, Normal Mode (8 Motors Working) & Dry Track

a. Case1: 0^0 Curve

Speed in kmph	Level Track	Slope 1/500	Slope 1/200	Slope 1/150	Slope 1/100	Slope 1/75	Slope 1/50	Slope 1/40
START	7600	7318	6471	6000	4651	3764	2704	2221
10	7600	7318	6471	6000	4651	3764	2704	2221
15	7600	7318	6471	6000	4651	3764	2704	2221
20	6000	6000	6000	6000	4651	3764	2704	2221
30	6000	6000	6000	6000	4651	3764	2704	2221
40	6000	6000	6000	6000	4651	3764	2660	2131
50	6000	6000	6000	6000	4651	3647	2488	1994
60	6000	6000	6000	6000	4312	3348	2288	1835
70	6000	6000	6000	4991	3601	2800	1913	1531
80	6000	6000	5194	4229	3064	2386	1629	1302
90	6000	6000	4443	3633	2643	2061	1407	1122
100	6000	6000	3840	3153	2304	1800	1228	977
110	6000	5332	3350	2763	2027	1586	1082	859
120	5626	4593	2931	2427	1789	1402	955	757

b. Case2: 1⁰ Curve

Speed in kmph	Level Track	Slope 1/500	Slope 1/200	Slope 1/150	Slope 1/100	Slope 1/75	Slope 1/50	Slope 1/40
START	7600	7318	6471	5844	4524	3679	2659	2189
10	7600	7318	6471	5844	4524	3679	2659	2189
15	7600	7318	6471	5844	4524	3679	2659	2189
20	6000	6000	6000	5844	4524	3679	2659	2189
30	6000	6000	6000	5844	4524	3679	2659	2189
40	6000	6000	6000	5844	4524	3679	2609	2097
50	6000	6000	6000	5844	4524	3550	2441	1963
60	6000	6000	6000	5734	4169	3259	2245	1805
70	6000	6000	5831	4773	3483	2726	1876	1506
80	6000	6000	4926	4047	2964	2323	1598	1281
90	6000	6000	4219	3479	2558	2007	1380	1104
100	6000	5745	3650	3022	2230	1753	1204	961
110	6000	4948	3189	2649	1963	1545	1061	845
120	5163	4275	2793	2329	1733	1365	936	744

c. Case3: 2⁰ Curve

Speed in kmph	Level Track	Slope 1/500	Slope 1/200	Slope 1/150	Slope 1/100	Slope 1/75	Slope 1/50	Slope 1/40
START	7600	7318	6471	5648	4404	3597	2614	2158
10	7600	7318	6471	5648	4404	3597	2614	2158
15	7600	7318	6471	5648	4404	3597	2614	2158
20	6000	6000	6000	5648	4404	3597	2614	2158
30	6000	6000	6000	5648	4404	3597	2614	2158
40	6000	6000	6000	5648	4404	3597	2559	2064
50	6000	6000	6000	5648	4404	3458	2395	1932
60	6000	6000	6000	5489	4035	3175	2203	1777
70	6000	6000	5538	4572	3372	2656	1841	1482
80	6000	6000	4683	3879	2870	2263	1568	1260
90	6000	6000	4015	3337	2478	1956	1353	1086
100	6000	5342	3478	2900	2161	1708	1181	945
110	5546	4614	3041	2545	1903	1505	1040	831
120	4769	3998	2667	2239	1680	1330	918	731

d. Case4: 3⁰ Curve

Speed in kmph	Level Track	Slope 1/500	Slope 1/200	Slope 1/150	Slope 1/100	Slope 1/75	Slope 1/50	Slope 1/40
START	7600	7318	6318	5464	4289	3519	2571	2128
10	7600	7318	6318	5464	4289	3519	2571	2128
15	7600	7318	6318	5464	4289	3519	2571	2128
20	6000	6000	6000	5464	4289	3519	2571	2128
30	6000	6000	6000	5464	4289	3519	2571	2128
40	6000	6000	6000	5464	4289	3519	2512	2032
50	6000	6000	6000	5464	4265	3370	2350	1902
60	6000	6000	6000	5264	3910	3095	2162	1749
70	6000	6000	5272	4387	3267	2589	1806	1459
80	6000	6000	4463	3724	2782	2206	1538	1240
90	6000	5820	3830	3206	2402	1907	1328	1068
100	5969	4991	3320	2788	2095	1665	1158	930
110	5133	4321	2906	2448	1845	1468	1020	817
120	4429	3753	2551	2155	1629	1297	900	719

e. Case5: 4⁰ Curve

Speed in kmph	Level Track	Slope 1/500	Slope 1/200	Slope 1/150	Slope 1/100	Slope 1/75	Slope 1/50	Slope 1/40
START	7600	7318	6090	5291	4180	3444	2530	2098
10	7600	7318	6090	5291	4180	3444	2530	2098
15	7600	7318	6090	5291	4180	3444	2530	2098
20	6000	6000	6000	5291	4180	3444	2530	2098
30	6000	6000	6000	5291	4180	3444	2530	2098
40	6000	6000	6000	5291	4180	3444	2466	2001
50	6000	6000	6000	5291	4135	3286	2307	1873
60	6000	6000	6000	5056	3791	3018	2122	1723
70	6000	6000	5029	4215	3169	2525	1773	1437
80	6000	6000	4261	3581	2699	2152	1510	1221
90	6000	5448	3660	3084	2330	1860	1303	1051
100	5537	4682	3176	2683	2033	1624	1137	915
110	4775	4062	2782	2357	1791	1432	1001	804
120	4132	3535	2444	2076	1582	1265	883	707

f. Case6: 5^0 Curve

Speed in kmph	Level Track	Slope 1/500	Slope 1/200	Slope 1/150	Slope 1/100	Slope 1/75	Slope 1/50	Slope 1/40
START	7600	7318	5878	5129	4077	3372	2489	2070
10	7600	7318	5878	5129	4077	3372	2489	2070
15	7600	7318	5878	5129	4077	3372	2489	2070
20	6000	6000	5878	5129	4077	3372	2489	2070
30	6000	6000	5878	5129	4077	3372	2489	2070
40	6000	6000	5878	5129	4077	3372	2422	1971
50	6000	6000	5878	5129	4012	3206	2266	1844
60	6000	6000	5777	4863	3679	2945	2084	1697
70	6000	6000	4808	4056	3076	2464	1741	1415
80	6000	6000	4076	3447	2620	2100	1482	1202
90	6000	5119	3504	2970	2263	1815	1279	1035
100	5161	4407	3043	2586	1975	1585	1116	900
110	4463	3831	2668	2273	1740	1397	982	791
120	3871	3340	2345	2003	1537	1235	867	696

8.2 Axle Weight 22.5 T, Degraded Mode (6 Motors Working) & Dry Track

a. Case1: 0^0 Curve

Speed in kmph	Level Track	Slope 1/500	Slope 1/200	Slope 1/150	Slope 1/100	Slope 1/75	Slope 1/50	Slope 1/40
START	7600	7318	5282	4489	3439	2774	1981	1618
10	7600	7318	5282	4489	3439	2774	1981	1618
15	7600	7318	5282	4489	3439	2774	1981	1618
20	6000	6000	5282	4489	3439	2774	1981	1618
30	6000	6000	5282	4489	3439	2774	1981	1618
40	6000	6000	5282	4489	3439	2774	1981	1618
50	6000	6000	5282	4489	3439	2774	1981	1618
60	6000	6000	5282	4489	3439	2774	1979	1582
70	6000	6000	5282	4343	3127	2426	1650	1316
80	6000	6000	4518	3674	2656	2063	1402	1115
90	6000	6000	3859	3151	2287	1778	1207	958
100	6000	5401	3330	2731	1989	1549	1050	831
110	5728	4622	2896	2384	1743	1359	919	725
120	4882	3981	2533	2093	1537	1199	810	637

b. Case2: 1^0 Curve

Speed in kmph	Level Track	Slope 1/500	Slope 1/200	Slope 1/150	Slope 1/100	Slope 1/75	Slope 1/50	Slope 1/40
START	7600	7251	5068	4332	3344	2711	1946	1594
10	7600	7251	5068	4332	3344	2711	1946	1594
15	7600	7251	5068	4332	3344	2711	1946	1594
20	6000	6000	5068	4332	3344	2711	1946	1594
30	6000	6000	5068	4332	3344	2711	1946	1594
40	6000	6000	5068	4332	3344	2711	1946	1594
50	6000	6000	5068	4332	3344	2711	1946	1594
60	6000	6000	5068	4332	3344	2711	1941	1557
70	6000	6000	5068	4152	3023	2361	1618	1295
80	6000	6000	4284	3515	2569	2008	1374	1097
90	6000	5863	3663	3017	2212	1731	1183	942
100	6000	4994	3165	2616	1925	1508	1029	816
110	5230	4288	2755	2285	1687	1323	901	713
120	4479	3704	2412	2008	1488	1167	793	626

c. Case3: 2⁰ Curve

Speed in kmph	Level Track	Slope 1/500	Slope 1/200	Slope 1/150	Slope 1/100	Slope 1/75	Slope 1/50	Slope 1/40
START	7600	6860	4871	4185	3253	2649	1913	1571
10	7600	6860	4871	4185	3253	2649	1913	1571
15	7600	6860	4871	4185	3253	2649	1913	1571
20	6000	6000	4871	4185	3253	2649	1913	1571
30	6000	6000	4871	4185	3253	2649	1913	1571
40	6000	6000	4871	4185	3253	2649	1913	1571
50	6000	6000	4871	4185	3253	2649	1913	1571
60	6000	6000	4871	4185	3253	2649	1904	1532
70	6000	6000	4821	3976	2926	2300	1587	1274
80	6000	6000	4071	3368	2487	1956	1348	1079
90	6000	5435	3486	2893	2142	1686	1160	926
100	5630	4643	3014	2510	1864	1468	1008	803
110	4809	3997	2627	2194	1635	1288	883	701
120	4135	3462	2302	1929	1441	1137	777	615

d. Case4: 3⁰ Curve

Speed in kmph	Level Track	Slope 1/500	Slope 1/200	Slope 1/150	Slope 1/100	Slope 1/75	Slope 1/50	Slope 1/40
START	7462	6509	4687	4048	3168	2591	1881	1549
10	7462	6509	4687	4048	3168	2591	1881	1549
15	7462	6509	4687	4048	3168	2591	1881	1549
20	6000	6000	4687	4048	3168	2591	1881	1549
30	6000	6000	4687	4048	3168	2591	1881	1549
40	6000	6000	4687	4048	3168	2591	1881	1549
50	6000	6000	4687	4048	3168	2591	1881	1549
60	6000	6000	4687	4048	3168	2591	1869	1508
70	6000	6000	4588	3814	2835	2241	1557	1254
80	6000	5973	3878	3233	2409	1906	1322	1061
90	6000	5063	3323	2778	2076	1643	1138	911
100	5190	4336	2877	2412	1807	1431	989	789
110	4449	3742	2509	2109	1585	1256	866	689
120	3838	3249	2201	1855	1398	1108	762	604

e. Case5: 4⁰ Curve

Speed in kmph	Level Track	Slope 1/500	Slope 1/200	Slope 1/150	Slope 1/100	Slope 1/75	Slope 1/50	Slope 1/40
START	7050	6191	4517	3918	3086	2535	1850	1527
10	7050	6191	4517	3918	3086	2535	1850	1527
15	7050	6191	4517	3918	3086	2535	1850	1527
20	6000	6000	4517	3918	3086	2535	1850	1527
30	6000	6000	4517	3918	3086	2535	1850	1527
40	6000	6000	4517	3918	3086	2535	1850	1527
50	6000	6000	4517	3918	3086	2535	1850	1527
60	6000	6000	4517	3918	3086	2535	1834	1484
70	6000	6000	4376	3664	2749	2186	1528	1234
80	6000	5577	3702	3108	2336	1858	1297	1044
90	5641	4737	3175	2672	2013	1602	1116	896
100	4812	4066	2751	2320	1753	1395	970	776
110	4138	3516	2401	2030	1537	1224	849	677
120	3580	3059	2108	1787	1356	1080	747	594

f. Case6: 5⁰ Curve

Speed in kmph	Level Track	Slope 1/500	Slope 1/200	Slope 1/150	Slope 1/100	Slope 1/75	Slope 1/50	Slope 1/40
START	6680	5902	4358	3797	3008	2481	1819	1505
10	6680	5902	4358	3797	3008	2481	1819	1505
15	6680	5902	4358	3797	3008	2481	1819	1505
20	6000	5902	4358	3797	3008	2481	1819	1505
30	6000	5902	4358	3797	3008	2481	1819	1505
40	6000	5902	4358	3797	3008	2481	1819	1505
50	6000	5902	4358	3797	3008	2481	1819	1505
60	6000	5902	4358	3797	3008	2481	1801	1462
70	6000	5902	4182	3525	2667	2132	1500	1215
80	6000	5229	3541	2991	2268	1813	1273	1028
90	5242	4450	3039	2572	1954	1563	1095	882
100	4484	3826	2634	2235	1702	1361	952	763
110	3865	3315	2301	1957	1493	1194	833	666
120	3352	2889	2022	1723	1317	1053	732	583

8.3 Axle Weight 22.5 T, Normal Mode (8 Motors Working) & Wet Track

a. Case1: 0^0 Curve

Speed in kmph	Level Track	Slope 1/500	Slope 1/200	Slope 1/150	Slope 1/100	Slope 1/75	Slope 1/50	Slope 1/40
START	7600	7318	5641	4796	3676	2968	2122	1736
10	7600	7318	5641	4796	3676	2968	2122	1736
15	7600	7318	5641	4796	3676	2968	2122	1736
20	6000	6000	5641	4796	3676	2968	2122	1736
30	6000	6000	5641	4796	3676	2968	2122	1736
40	6000	6000	5641	4796	3676	2968	2122	1736
50	6000	6000	5641	4796	3676	2968	2122	1736
60	6000	6000	5641	4796	3676	2968	2122	1736
70	6000	6000	5641	4796	3601	2800	1913	1531
80	6000	6000	5194	4229	3064	2386	1629	1302
90	6000	6000	4443	3633	2643	2061	1407	1122
100	6000	6000	3840	3153	2304	1800	1228	977
110	6000	5332	3350	2763	2027	1586	1082	859
120	5626	4593	2931	2427	1789	1402	955	757

b. Case2: 1^0 Curve

Speed in kmph	Level Track	Slope 1/500	Slope 1/200	Slope 1/150	Slope 1/100	Slope 1/75	Slope 1/50	Slope 1/40
START	7600	7318	5413	4628	3575	2900	2086	1711
10	7600	7318	5413	4628	3575	2900	2086	1711
15	7600	7318	5413	4628	3575	2900	2086	1711
20	6000	6000	5413	4628	3575	2900	2086	1711
30	6000	6000	5413	4628	3575	2900	2086	1711
40	6000	6000	5413	4628	3575	2900	2086	1711
50	6000	6000	5413	4628	3575	2900	2086	1711
60	6000	6000	5413	4628	3575	2900	2086	1711
70	6000	6000	5413	4628	3483	2726	1876	1506
80	6000	6000	4926	4047	2964	2323	1598	1281
90	6000	6000	4219	3479	2558	2007	1380	1104
100	6000	5745	3650	3022	2230	1753	1204	961
110	6000	4948	3189	2649	1963	1545	1061	845
120	5163	4275	2793	2329	1733	1365	936	744

c. Case3: 2⁰ Curve

Speed in kmph	Level Track	Slope 1/500	Slope 1/200	Slope 1/150	Slope 1/100	Slope 1/75	Slope 1/50	Slope 1/40
START	7600	7318	5202	4472	3479	2835	2050	1686
10	7600	7318	5202	4472	3479	2835	2050	1686
15	7600	7318	5202	4472	3479	2835	2050	1686
20	6000	6000	5202	4472	3479	2835	2050	1686
30	6000	6000	5202	4472	3479	2835	2050	1686
40	6000	6000	5202	4472	3479	2835	2050	1686
50	6000	6000	5202	4472	3479	2835	2050	1686
60	6000	6000	5202	4472	3479	2835	2050	1686
70	6000	6000	5202	4472	3372	2656	1841	1482
80	6000	6000	4683	3879	2870	2263	1568	1260
90	6000	6000	4015	3337	2478	1956	1353	1086
100	6000	5342	3478	2900	2161	1708	1181	945
110	5546	4614	3041	2545	1903	1505	1040	831
120	4769	3998	2667	2239	1680	1330	918	731

d. Case4: 3⁰ Curve

Speed in kmph	Level Track	Slope 1/500	Slope 1/200	Slope 1/150	Slope 1/100	Slope 1/75	Slope 1/50	Slope 1/40
START	7600	6948	5007	4325	3387	2773	2016	1662
10	7600	6948	5007	4325	3387	2773	2016	1662
15	7600	6948	5007	4325	3387	2773	2016	1662
20	6000	6000	5007	4325	3387	2773	2016	1662
30	6000	6000	5007	4325	3387	2773	2016	1662
40	6000	6000	5007	4325	3387	2773	2016	1662
50	6000	6000	5007	4325	3387	2773	2016	1662
60	6000	6000	5007	4325	3387	2773	2016	1662
70	6000	6000	5007	4325	3267	2589	1806	1459
80	6000	6000	4463	3724	2782	2206	1538	1240
90	6000	5820	3830	3206	2402	1907	1328	1068
100	5969	4991	3320	2788	2095	1665	1158	930
110	5133	4321	2906	2448	1845	1468	1020	817
120	4429	3753	2551	2155	1629	1297	900	719

e. Case5: 4^0 Curve

Speed in kmph	Level Track	Slope 1/500	Slope 1/200	Slope 1/150	Slope 1/100	Slope 1/75	Slope 1/50	Slope 1/40
START	7524	6609	4825	4187	3300	2713	1983	1639
10	7524	6609	4825	4187	3300	2713	1983	1639
15	7524	6609	4825	4187	3300	2713	1983	1639
20	6000	6000	4825	4187	3300	2713	1983	1639
30	6000	6000	4825	4187	3300	2713	1983	1639
40	6000	6000	4825	4187	3300	2713	1983	1639
50	6000	6000	4825	4187	3300	2713	1983	1639
60	6000	6000	4825	4187	3300	2713	1983	1639
70	6000	6000	4825	4187	3169	2525	1773	1437
80	6000	6000	4261	3581	2699	2152	1510	1221
90	6000	5448	3660	3084	2330	1860	1303	1051
100	5537	4682	3176	2683	2033	1624	1137	915
110	4775	4062	2782	2357	1791	1432	1001	804
120	4132	3535	2444	2076	1582	1265	883	707

f. Case6: 5^0 Curve

Speed in kmph	Level Track	Slope 1/500	Slope 1/200	Slope 1/150	Slope 1/100	Slope 1/75	Slope 1/50	Slope 1/40
START	7130	6301	4655	4058	3218	2656	1951	1616
10	7130	6301	4655	4058	3218	2656	1951	1616
15	7130	6301	4655	4058	3218	2656	1951	1616
20	6000	6000	4655	4058	3218	2656	1951	1616
30	6000	6000	4655	4058	3218	2656	1951	1616
40	6000	6000	4655	4058	3218	2656	1951	1616
50	6000	6000	4655	4058	3218	2656	1951	1616
60	6000	6000	4655	4058	3218	2656	1951	1616
70	6000	6000	4655	4056	3076	2464	1741	1415
80	6000	6000	4076	3447	2620	2100	1482	1202
90	6000	5119	3504	2970	2263	1815	1279	1035
100	5161	4407	3043	2586	1975	1585	1116	900
110	4463	3831	2668	2273	1740	1397	982	791
120	3871	3340	2345	2003	1537	1235	867	696

8.4 Axle Weight 22.5 T, Degraded Mode (6 Motors Working) & Wet Track

a. Case1: 0^0 Curve

Speed in kmph	Level Track	Slope 1/500	Slope 1/200	Slope 1/150	Slope 1/100	Slope 1/75	Slope 1/50	Slope 1/40
START	7173	6094	4176	3544	2706	2176	1543	1254
10	7173	6094	4176	3544	2706	2176	1543	1254
15	7173	6094	4176	3544	2706	2176	1543	1254
20	6000	6000	4176	3544	2706	2176	1543	1254
30	6000	6000	4176	3544	2706	2176	1543	1254
40	6000	6000	4176	3544	2706	2176	1543	1254
50	6000	6000	4176	3544	2706	2176	1543	1254
60	6000	6000	4176	3544	2706	2176	1543	1254
70	6000	6000	4176	3544	2706	2176	1543	1254
80	6000	6000	4176	3544	2656	2063	1402	1115
90	6000	6000	3859	3151	2287	1778	1207	958
100	6000	5401	3330	2731	1989	1549	1050	831
110	5728	4622	2896	2384	1743	1359	919	725
120	4882	3981	2533	2093	1537	1199	810	637

b. Case2: 1^0 Curve

Speed in kmph	Level Track	Slope 1/500	Slope 1/200	Slope 1/150	Slope 1/100	Slope 1/75	Slope 1/50	Slope 1/40
START	6699	5746	4006	3419	2630	2125	1516	1235
10	6699	5746	4006	3419	2630	2125	1516	1235
15	6699	5746	4006	3419	2630	2125	1516	1235
20	6000	5746	4006	3419	2630	2125	1516	1235
30	6000	5746	4006	3419	2630	2125	1516	1235
40	6000	5746	4006	3419	2630	2125	1516	1235
50	6000	5746	4006	3419	2630	2125	1516	1235
60	6000	5746	4006	3419	2630	2125	1516	1235
70	6000	5746	4006	3419	2630	2125	1516	1235
80	6000	5746	4006	3419	2569	2008	1374	1097
90	6000	5746	3663	3017	2212	1731	1183	942
100	6000	4994	3165	2616	1925	1508	1029	816
110	5230	4288	2755	2285	1687	1323	901	713
120	4479	3704	2412	2008	1488	1167	793	626

c. Case3: 2⁰ Curve

Speed in kmph	Level Track	Slope 1/500	Slope 1/200	Slope 1/150	Slope 1/100	Slope 1/75	Slope 1/50	Slope 1/40
START	6284	5435	3848	3301	2558	2077	1489	1217
10	6284	5435	3848	3301	2558	2077	1489	1217
15	6284	5435	3848	3301	2558	2077	1489	1217
20	6000	5435	3848	3301	2558	2077	1489	1217
30	6000	5435	3848	3301	2558	2077	1489	1217
40	6000	5435	3848	3301	2558	2077	1489	1217
50	6000	5435	3848	3301	2558	2077	1489	1217
60	6000	5435	3848	3301	2558	2077	1489	1217
70	6000	5435	3848	3301	2558	2077	1489	1217
80	6000	5435	3848	3301	2487	1956	1348	1079
90	6000	5435	3486	2893	2142	1686	1160	926
100	5630	4643	3014	2510	1864	1468	1008	803
110	4809	3997	2627	2194	1635	1288	883	701
120	4135	3462	2302	1929	1441	1137	777	615

d. Case4: 3⁰ Curve

Speed in kmph	Level Track	Slope 1/500	Slope 1/200	Slope 1/150	Slope 1/100	Slope 1/75	Slope 1/50	Slope 1/40
START	5915	5155	3702	3192	2490	2030	1464	1199
10	5915	5155	3702	3192	2490	2030	1464	1199
15	5915	5155	3702	3192	2490	2030	1464	1199
20	5915	5155	3702	3192	2490	2030	1464	1199
30	5915	5155	3702	3192	2490	2030	1464	1199
40	5915	5155	3702	3192	2490	2030	1464	1199
50	5915	5155	3702	3192	2490	2030	1464	1199
60	5915	5155	3702	3192	2490	2030	1464	1199
70	5915	5155	3702	3192	2490	2030	1464	1199
80	5915	5155	3702	3192	2409	1906	1322	1061
90	5915	5063	3323	2778	2076	1643	1138	911
100	5190	4336	2877	2412	1807	1431	989	789
110	4449	3742	2509	2109	1585	1256	866	689
120	3838	3249	2201	1855	1398	1108	762	604

e. Case5: 4⁰ Curve

Speed in kmph	Level Track	Slope 1/500	Slope 1/200	Slope 1/150	Slope 1/100	Slope 1/75	Slope 1/50	Slope 1/40
START	5586	4901	3566	3089	2425	1985	1439	1181
10	5586	4901	3566	3089	2425	1985	1439	1181
15	5586	4901	3566	3089	2425	1985	1439	1181
20	5586	4901	3566	3089	2425	1985	1439	1181
30	5586	4901	3566	3089	2425	1985	1439	1181
40	5586	4901	3566	3089	2425	1985	1439	1181
50	5586	4901	3566	3089	2425	1985	1439	1181
60	5586	4901	3566	3089	2425	1985	1439	1181
70	5586	4901	3566	3089	2425	1985	1439	1181
80	5586	4901	3566	3089	2336	1858	1297	1044
90	5586	4737	3175	2672	2013	1602	1116	896
100	4812	4066	2751	2320	1753	1395	970	776
110	4138	3516	2401	2030	1537	1224	849	677
120	3580	3059	2108	1787	1356	1080	747	594

f. Case6: 5⁰ Curve

Speed in kmph	Level Track	Slope 1/500	Slope 1/200	Slope 1/150	Slope 1/100	Slope 1/75	Slope 1/50	Slope 1/40
START	5291	4670	3439	2992	2363	1942	1415	1164
10	5291	4670	3439	2992	2363	1942	1415	1164
15	5291	4670	3439	2992	2363	1942	1415	1164
20	5291	4670	3439	2992	2363	1942	1415	1164
30	5291	4670	3439	2992	2363	1942	1415	1164
40	5291	4670	3439	2992	2363	1942	1415	1164
50	5291	4670	3439	2992	2363	1942	1415	1164
60	5291	4670	3439	2992	2363	1942	1415	1164
70	5291	4670	3439	2992	2363	1942	1415	1164
80	5291	4670	3439	2991	2268	1813	1273	1028
90	5242	4450	3039	2572	1954	1563	1095	882
100	4484	3826	2634	2235	1702	1361	952	763
110	3865	3315	2301	1957	1493	1194	833	666
120	3352	2889	2022	1723	1317	1053	732	583

8.5 Axle Weight 25 T, Normal Mode (8 Motors Working) & Dry Track

a. Case1: 0^0 Curve

Speed in kmph	Level Track	Slope 1/500	Slope 1/200	Slope 1/150	Slope 1/100	Slope 1/75	Slope 1/50	Slope 1/40
START	7600	7371	6682	6300	5174	4187	3008	2470
10	7600	7371	6682	6300	5174	4187	3008	2470
15	7600	7371	6682	6300	5174	4187	3008	2470
20	6300	6300	6300	6300	5174	4187	3008	2470
30	6300	6300	6300	6300	5174	4187	3008	2418
40	6300	6300	6300	6300	5174	4146	2791	2224
50	6300	6300	6300	6300	4988	3826	2581	2057
60	6300	6300	6300	6300	4468	3436	2321	1849
70	6300	6300	6300	5248	3727	2870	1937	1540
80	6300	6300	5521	4440	3167	2442	1647	1307
90	6300	6300	4712	3808	2729	2108	1420	1124
100	6300	6300	4064	3299	2375	1838	1237	977
110	6300	5833	3538	2886	2088	1617	1088	857
120	6252	5002	3089	2531	1839	1427	959	753

b. Case2: 1^0 Curve

Speed in kmph	Level Track	Slope 1/500	Slope 1/200	Slope 1/150	Slope 1/100	Slope 1/75	Slope 1/50	Slope 1/40
START	7600	7371	6682	6300	5033	4092	2958	2435
10	7600	7371	6682	6300	5033	4092	2958	2435
15	7600	7371	6682	6300	5033	4092	2958	2435
20	6300	6300	6300	6300	5033	4092	2958	2435
30	6300	6300	6300	6300	5033	4092	2958	2379
40	6300	6300	6300	6300	5033	4031	2736	2188
50	6300	6300	6300	6300	4814	3721	2530	2024
60	6300	6300	6300	6023	4314	3342	2275	1819
70	6300	6300	6188	5006	3600	2791	1899	1515
80	6300	6300	5218	4238	3060	2376	1615	1285
90	6300	6300	4460	3638	2637	2050	1392	1105
100	6300	6269	3851	3155	2296	1788	1212	960
110	6300	5377	3358	2762	2019	1574	1066	842
120	5686	4627	2935	2424	1779	1389	939	740

c. Case3: 2⁰ Curve

Speed in kmph	Level Track	Slope 1/500	Slope 1/200	Slope 1/150	Slope 1/100	Slope 1/75	Slope 1/50	Slope 1/40
START	7600	7371	6682	6283	4899	4002	2908	2401
10	7600	7371	6682	6283	4899	4002	2908	2401
15	7600	7371	6682	6283	4899	4002	2908	2401
20	6300	6300	6300	6283	4899	4002	2908	2401
30	6300	6300	6300	6283	4899	4002	2908	2341
40	6300	6300	6300	6283	4899	3921	2683	2153
50	6300	6300	6300	6283	4652	3621	2482	1991
60	6300	6300	6300	5753	4170	3252	2231	1790
70	6300	6300	5858	4784	3480	2717	1862	1490
80	6300	6300	4945	4054	2959	2313	1583	1264
90	6300	6300	4232	3482	2551	1996	1364	1087
100	6300	5792	3659	3022	2222	1740	1188	944
110	6091	4985	3194	2647	1954	1532	1045	828
120	5211	4303	2795	2325	1722	1352	920	727

d. Case4: 3⁰ Curve

Speed in kmph	Level Track	Slope 1/500	Slope 1/200	Slope 1/150	Slope 1/100	Slope 1/75	Slope 1/50	Slope 1/40
START	7600	7371	6682	6078	4771	3915	2860	2367
10	7600	7371	6682	6078	4771	3915	2860	2367
15	7600	7371	6682	6078	4771	3915	2860	2367
20	6300	6300	6300	6078	4771	3915	2860	2367
30	6300	6300	6300	6078	4771	3915	2860	2304
40	6300	6300	6300	6078	4771	3818	2632	2119
50	6300	6300	6300	6078	4500	3526	2434	1960
60	6300	6300	6300	5505	4035	3168	2189	1761
70	6300	6300	5560	4581	3368	2646	1826	1466
80	6300	6300	4699	3884	2865	2252	1552	1243
90	6300	6296	4025	3338	2470	1944	1338	1069
100	6300	5381	3484	2899	2152	1695	1165	928
110	5596	4644	3044	2542	1893	1492	1024	813
120	4807	4020	2667	2234	1669	1317	902	714

e. Case5: 4⁰ Curve

Speed in kmph	Level Track	Slope 1/500	Slope 1/200	Slope 1/150	Slope 1/100	Slope 1/75	Slope 1/50	Slope 1/40
START	7600	7371	6682	5886	4650	3832	2814	2334
10	7600	7371	6682	5886	4650	3832	2814	2334
15	7600	7371	6682	5886	4650	3832	2814	2334
20	6300	6300	6300	5886	4650	3832	2814	2334
30	6300	6300	6300	5886	4650	3832	2808	2268
40	6300	6300	6300	5886	4650	3719	2583	2085
50	6300	6300	6300	5886	4357	3436	2389	1929
60	6300	6300	6300	5277	3908	3087	2148	1734
70	6300	6300	5290	4394	3263	2579	1792	1443
80	6300	6300	4475	3727	2776	2195	1523	1223
90	6300	5862	3837	3205	2394	1894	1312	1051
100	6021	5022	3324	2785	2086	1652	1142	913
110	5174	4345	2908	2443	1835	1454	1004	800
120	4460	3770	2550	2149	1618	1283	884	702

f. Case6: 5⁰ Curve

Speed in kmph	Level Track	Slope 1/500	Slope 1/200	Slope 1/150	Slope 1/100	Slope 1/75	Slope 1/50	Slope 1/40
START	7600	7371	6539	5706	4535	3752	2769	2303
10	7600	7371	6539	5706	4535	3752	2769	2303
15	7600	7371	6539	5706	4535	3752	2769	2303
20	6300	6300	6300	5706	4535	3752	2769	2303
30	6300	6300	6300	5706	4535	3752	2756	2233
40	6300	6300	6300	5706	4535	3626	2535	2053
50	6300	6300	6300	5665	4222	3350	2345	1899
60	6300	6300	6070	5066	3789	3010	2108	1707
70	6300	6300	5045	4220	3163	2514	1758	1420
80	6300	6300	4271	3582	2691	2140	1494	1204
90	6300	5482	3665	3082	2322	1847	1287	1034
100	5579	4707	3178	2679	2023	1611	1121	898
110	4808	4081	2782	2352	1780	1418	985	786
120	4157	3548	2441	2069	1570	1251	867	690

8.6 Axle Weight 25 T, Degraded Mode (6 Motors Working) & Dry Track

a. Case1: 0⁰ Curve

Speed in kmph	Level Track	Slope 1/500	Slope 1/200	Slope 1/150	Slope 1/100	Slope 1/75	Slope 1/50	Slope 1/40
START	7600	7371	5874	4992	3824	3085	2202	1799
10	7600	7371	5874	4992	3824	3085	2202	1799
15	7600	7371	5874	4992	3824	3085	2202	1799
20	6300	6300	5874	4992	3824	3085	2202	1799
30	6300	6300	5874	4992	3824	3085	2202	1799
40	6300	6300	5874	4992	3824	3085	2202	1765
50	6300	6300	5874	4992	3824	3081	2087	1664
60	6300	6300	5874	4992	3730	2886	1960	1563
70	6300	6300	5343	4324	3107	2407	1631	1297
80	6300	6300	4500	3656	2637	2044	1382	1096
90	6300	6300	3841	3133	2268	1759	1187	938
100	6300	5384	3313	2712	1970	1530	1030	811
110	5712	4606	2878	2366	1725	1340	900	706
120	4867	3965	2515	2075	1518	1180	791	618

b. Case2: 1⁰ Curve

Speed in kmph	Level Track	Slope 1/500	Slope 1/200	Slope 1/150	Slope 1/100	Slope 1/75	Slope 1/50	Slope 1/40
START	7600	7371	5636	4817	3718	3014	2164	1773
10	7600	7371	5636	4817	3718	3014	2164	1773
15	7600	7371	5636	4817	3718	3014	2164	1773
20	6300	6300	5636	4817	3718	3014	2164	1773
30	6300	6300	5636	4817	3718	3014	2164	1773
40	6300	6300	5636	4817	3718	3014	2164	1736
50	6300	6300	5636	4817	3718	2998	2046	1637
60	6300	6300	5636	4817	3605	2809	1921	1537
70	6300	6300	5058	4133	3004	2342	1599	1275
80	6300	6300	4265	3496	2549	1989	1355	1077
90	6300	5846	3645	2998	2193	1712	1164	922
100	6131	4977	3147	2598	1906	1489	1009	797
110	5214	4271	2737	2267	1669	1304	882	694
120	4463	3688	2395	1990	1469	1149	774	606

c. Case3: 2⁰ Curve

Speed in kmph	Level Track	Slope 1/500	Slope 1/200	Slope 1/150	Slope 1/100	Slope 1/75	Slope 1/50	Slope 1/40
START	7600	7371	5416	4654	3618	2946	2127	1747
10	7600	7371	5416	4654	3618	2946	2127	1747
15	7600	7371	5416	4654	3618	2946	2127	1747
20	6300	6300	5416	4654	3618	2946	2127	1747
30	6300	6300	5416	4654	3618	2946	2127	1747
40	6300	6300	5416	4654	3618	2946	2127	1708
50	6300	6300	5416	4654	3618	2918	2007	1610
60	6300	6300	5416	4654	3488	2735	1884	1512
70	6300	6300	4802	3957	2907	2281	1568	1254
80	6300	6300	4053	3350	2467	1936	1328	1059
90	6300	5417	3467	2874	2123	1667	1140	906
100	5613	4625	2996	2492	1845	1449	989	783
110	4793	3980	2609	2176	1616	1270	864	681
120	4119	3446	2285	1911	1423	1118	758	595

d. Case4: 3⁰ Curve

Speed in kmph	Level Track	Slope 1/500	Slope 1/200	Slope 1/150	Slope 1/100	Slope 1/75	Slope 1/50	Slope 1/40
START	7600	7238	5212	4501	3522	2881	2092	1722
10	7600	7238	5212	4501	3522	2881	2092	1722
15	7600	7238	5212	4501	3522	2881	2092	1722
20	6300	6300	5212	4501	3522	2881	2092	1722
30	6300	6300	5212	4501	3522	2881	2092	1722
40	6300	6300	5212	4501	3522	2881	2089	1681
50	6300	6300	5212	4501	3522	2843	1969	1585
60	6300	6300	5212	4501	3378	2665	1849	1488
70	6300	6300	4569	3795	2816	2222	1538	1234
80	6300	5955	3860	3214	2390	1887	1302	1042
90	6085	5045	3305	2759	2057	1624	1118	891
100	5173	4318	2859	2393	1788	1412	970	770
110	4433	3725	2491	2091	1566	1237	847	669
120	3822	3232	2183	1837	1379	1089	743	585

e. Case5: 4⁰ Curve

Speed in kmph	Level Track	Slope 1/500	Slope 1/200	Slope 1/150	Slope 1/100	Slope 1/75	Slope 1/50	Slope 1/40
START	7600	6884	5022	4357	3432	2819	2057	1698
10	7600	6884	5022	4357	3432	2819	2057	1698
15	7600	6884	5022	4357	3432	2819	2057	1698
20	6300	6300	5022	4357	3432	2819	2057	1698
30	6300	6300	5022	4357	3432	2819	2057	1698
40	6300	6300	5022	4357	3432	2819	2050	1655
50	6300	6300	5022	4357	3432	2771	1932	1560
60	6300	6300	5022	4357	3274	2598	1814	1465
70	6300	6300	4357	3645	2729	2166	1509	1214
80	6300	5559	3683	3089	2317	1839	1278	1025
90	5623	4719	3156	2653	1994	1583	1097	876
100	4795	4048	2732	2302	1734	1376	951	757
110	4121	3499	2383	2012	1519	1205	830	658
120	3563	3042	2090	1769	1338	1061	728	574

f. Case6: 5⁰ Curve

Speed in kmph	Level Track	Slope 1/500	Slope 1/200	Slope 1/150	Slope 1/100	Slope 1/75	Slope 1/50	Slope 1/40
START	7428	6563	4846	4222	3345	2759	2023	1674
10	7428	6563	4846	4222	3345	2759	2023	1674
15	7428	6563	4846	4222	3345	2759	2023	1674
20	6300	6300	4846	4222	3345	2759	2023	1674
30	6300	6300	4846	4222	3345	2759	2023	1674
40	6300	6300	4846	4222	3345	2759	2012	1629
50	6300	6300	4846	4222	3345	2703	1897	1535
60	6300	6300	4846	4212	3176	2534	1781	1442
70	6300	6209	4163	3506	2648	2113	1481	1195
80	6174	5210	3522	2972	2248	1794	1254	1008
90	5225	4432	3020	2554	1935	1544	1076	862
100	4467	3808	2616	2217	1683	1342	932	744
110	3849	3297	2283	1939	1474	1175	813	646
120	3336	2872	2004	1705	1298	1035	713	564

8.7 Axle Weight 25 T, Normal Mode (8 Motors Working) & Wet Track

a. Case1: 0^0 Curve

Speed in kmph	Level Track	Slope 1/500	Slope 1/200	Slope 1/150	Slope 1/100	Slope 1/75	Slope 1/50	Slope 1/40
START	7600	7371	6271	5332	4087	3300	2359	1930
10	7600	7371	6271	5332	4087	3300	2359	1930
15	7600	7371	6271	5332	4087	3300	2359	1930
20	6300	6300	6271	5332	4087	3300	2359	1930
30	6300	6300	6271	5332	4087	3300	2359	1930
40	6300	6300	6271	5332	4087	3300	2359	1930
50	6300	6300	6271	5332	4087	3300	2359	1930
60	6300	6300	6271	5332	4087	3300	2269	1815
70	6300	6300	6125	4962	3574	2775	1889	1508
80	6300	6300	5176	4210	3045	2366	1610	1282
90	6300	6300	4425	3614	2624	2042	1388	1103
100	6300	6194	3822	3135	2285	1781	1209	958
110	6300	5316	3333	2745	2009	1567	1063	840
120	5611	4577	2914	2409	1771	1383	936	738

b. Case2: 1^0 Curve

Speed in kmph	Level Track	Slope 1/500	Slope 1/200	Slope 1/150	Slope 1/100	Slope 1/75	Slope 1/50	Slope 1/40
START	7600	7371	6017	5145	3974	3224	2319	1902
10	7600	7371	6017	5145	3974	3224	2319	1902
15	7600	7371	6017	5145	3974	3224	2319	1902
20	6300	6300	6017	5145	3974	3224	2319	1902
30	6300	6300	6017	5145	3974	3224	2319	1902
40	6300	6300	6017	5145	3974	3224	2319	1902
50	6300	6300	6017	5145	3974	3224	2319	1902
60	6300	6300	6017	5145	3974	3224	2225	1786
70	6300	6300	5801	4744	3457	2701	1853	1483
80	6300	6300	4907	4028	2945	2304	1578	1261
90	6300	6300	4200	3460	2539	1988	1360	1084
100	6300	5728	3632	3003	2211	1734	1185	942
110	6013	4932	3171	2631	1945	1526	1041	826
120	5148	4259	2776	2312	1714	1347	917	725

c. Case3: 2^0 Curve

Speed in kmph	Level Track	Slope 1/500	Slope 1/200	Slope 1/150	Slope 1/100	Slope 1/75	Slope 1/50	Slope 1/40
START	7600	7371	5783	4971	3867	3152	2280	1875
10	7600	7371	5783	4971	3867	3152	2280	1875
15	7600	7371	5783	4971	3867	3152	2280	1875
20	6300	6300	5783	4971	3867	3152	2280	1875
30	6300	6300	5783	4971	3867	3152	2280	1875
40	6300	6300	5783	4971	3867	3152	2280	1875
50	6300	6300	5783	4971	3867	3152	2280	1875
60	6300	6300	5783	4971	3867	3152	2183	1757
70	6300	6300	5508	4544	3346	2631	1817	1460
80	6300	6300	4665	3860	2851	2244	1548	1240
90	6300	6228	3997	3318	2459	1937	1334	1066
100	6300	5325	3460	2882	2142	1689	1162	926
110	5530	4597	3024	2527	1884	1487	1021	811
120	4753	3981	2649	2221	1661	1312	899	712

d. Case4: 3^0 Curve

Speed in kmph	Level Track	Slope 1/500	Slope 1/200	Slope 1/150	Slope 1/100	Slope 1/75	Slope 1/50	Slope 1/40
START	7600	7371	5566	4808	3766	3083	2241	1848
10	7600	7371	5566	4808	3766	3083	2241	1848
15	7600	7371	5566	4808	3766	3083	2241	1848
20	6300	6300	5566	4808	3766	3083	2241	1848
30	6300	6300	5566	4808	3766	3083	2241	1848
40	6300	6300	5566	4808	3766	3083	2241	1848
50	6300	6300	5566	4808	3766	3083	2241	1848
60	6300	6300	5566	4808	3766	3075	2142	1730
70	6300	6300	5243	4359	3241	2564	1783	1436
80	6300	6300	4444	3705	2763	2187	1519	1220
90	6300	5803	3811	3187	2383	1887	1309	1049
100	5952	4974	3302	2769	2076	1646	1139	910
110	5117	4304	2888	2429	1827	1449	1001	798
120	4413	3736	2533	2137	1611	1278	881	700

e. Case5: 4⁰ Curve

Speed in kmph	Level Track	Slope 1/500	Slope 1/200	Slope 1/150	Slope 1/100	Slope 1/75	Slope 1/50	Slope 1/40
START	7600	7347	5364	4655	3669	3016	2204	1822
10	7600	7347	5364	4655	3669	3016	2204	1822
15	7600	7347	5364	4655	3669	3016	2204	1822
20	6300	6300	5364	4655	3669	3016	2204	1822
30	6300	6300	5364	4655	3669	3016	2204	1822
40	6300	6300	5364	4655	3669	3016	2204	1822
50	6300	6300	5364	4655	3669	3016	2204	1822
60	6300	6300	5364	4655	3669	2999	2103	1703
70	6300	6300	5001	4188	3143	2500	1750	1414
80	6300	6300	4242	3562	2679	2132	1490	1201
90	6300	5430	3641	3065	2311	1840	1284	1032
100	5519	4664	3158	2665	2014	1605	1117	895
110	4759	4045	2764	2339	1773	1413	982	784
120	4116	3518	2426	2058	1563	1247	864	688

f. Case6: 5⁰ Curve

Speed in kmph	Level Track	Slope 1/500	Slope 1/200	Slope 1/150	Slope 1/100	Slope 1/75	Slope 1/50	Slope 1/40
START	7600	7005	5176	4511	3577	2952	2169	1796
10	7600	7005	5176	4511	3577	2952	2169	1796
15	7600	7005	5176	4511	3577	2952	2169	1796
20	6300	6300	5176	4511	3577	2952	2169	1796
30	6300	6300	5176	4511	3577	2952	2169	1796
40	6300	6300	5176	4511	3577	2952	2169	1796
50	6300	6300	5176	4511	3577	2952	2169	1796
60	6300	6300	5176	4511	3577	2926	2065	1677
70	6300	6300	4779	4029	3050	2439	1718	1392
80	6300	5989	4057	3428	2601	2080	1463	1182
90	6008	5101	3485	2951	2244	1795	1260	1015
100	5144	4389	3024	2567	1956	1566	1096	881
110	4446	3814	2650	2254	1721	1378	963	771
120	3855	3323	2327	1984	1518	1216	847	676

8.8 Axle Weight 25 T, Degraded Mode (6 Motors Working) & Wet Track

a. Case1: 0^0 Curve

Speed in kmph	Level Track	Slope 1/500	Slope 1/200	Slope 1/150	Slope 1/100	Slope 1/75	Slope 1/50	Slope 1/40
START	7600	6771	4640	3938	3007	2418	1715	1393
10	7600	6771	4640	3938	3007	2418	1715	1393
15	7600	6771	4640	3938	3007	2418	1715	1393
20	6300	6300	4640	3938	3007	2418	1715	1393
30	6300	6300	4640	3938	3007	2418	1715	1393
40	6300	6300	4640	3938	3007	2418	1715	1393
50	6300	6300	4640	3938	3007	2418	1715	1393
60	6300	6300	4640	3938	3007	2418	1715	1393
70	6300	6300	4640	3938	3007	2406	1630	1296
80	6300	6300	4498	3654	2636	2043	1382	1095
90	6300	6300	3840	3132	2267	1759	1187	938
100	6300	5382	3311	2711	1970	1529	1030	811
110	5710	4604	2877	2365	1724	1339	900	706
120	4865	3963	2514	2074	1517	1180	790	617

b. Case2: 1^0 Curve

Speed in kmph	Level Track	Slope 1/500	Slope 1/200	Slope 1/150	Slope 1/100	Slope 1/75	Slope 1/50	Slope 1/40
START	7444	6384	4451	3798	2922	2361	1684	1372
10	7444	6384	4451	3798	2922	2361	1684	1372
15	7444	6384	4451	3798	2922	2361	1684	1372
20	6300	6300	4451	3798	2922	2361	1684	1372
30	6300	6300	4451	3798	2922	2361	1684	1372
40	6300	6300	4451	3798	2922	2361	1684	1372
50	6300	6300	4451	3798	2922	2361	1684	1372
60	6300	6300	4451	3798	2922	2361	1684	1372
70	6300	6300	4451	3798	2922	2341	1598	1275
80	6300	6300	4264	3495	2549	1988	1354	1077
90	6300	5843	3644	2997	2192	1711	1163	922
100	6129	4975	3146	2596	1905	1488	1009	797
110	5211	4269	2736	2266	1668	1303	881	693
120	4461	3686	2394	1989	1468	1148	774	606

c. Case3: 2⁰ Curve

Speed in kmph	Level Track	Slope 1/500	Slope 1/200	Slope 1/150	Slope 1/100	Slope 1/75	Slope 1/50	Slope 1/40
START	6982	6039	4276	3668	2843	2307	1655	1352
10	6982	6039	4276	3668	2843	2307	1655	1352
15	6982	6039	4276	3668	2843	2307	1655	1352
20	6300	6039	4276	3668	2843	2307	1655	1352
30	6300	6039	4276	3668	2843	2307	1655	1352
40	6300	6039	4276	3668	2843	2307	1655	1352
50	6300	6039	4276	3668	2843	2307	1655	1352
60	6300	6039	4276	3668	2843	2307	1655	1352
70	6300	6039	4276	3668	2843	2280	1567	1254
80	6300	6039	4051	3348	2467	1936	1328	1059
90	6300	5415	3466	2873	2122	1666	1140	906
100	5611	4623	2995	2490	1845	1449	989	783
110	4791	3978	2608	2175	1615	1269	863	681
120	4117	3444	2283	1910	1422	1118	758	595

d. Case4: 3⁰ Curve

Speed in kmph	Level Track	Slope 1/500	Slope 1/200	Slope 1/150	Slope 1/100	Slope 1/75	Slope 1/50	Slope 1/40
START	6572	5727	4113	3546	2767	2255	1626	1332
10	6572	5727	4113	3546	2767	2255	1626	1332
15	6572	5727	4113	3546	2767	2255	1626	1332
20	6300	5727	4113	3546	2767	2255	1626	1332
30	6300	5727	4113	3546	2767	2255	1626	1332
40	6300	5727	4113	3546	2767	2255	1626	1332
50	6300	5727	4113	3546	2767	2255	1626	1332
60	6300	5727	4113	3546	2767	2255	1626	1332
70	6300	5727	4113	3546	2767	2221	1537	1233
80	6300	5727	3858	3213	2389	1886	1302	1041
90	6083	5043	3304	2758	2056	1623	1118	891
100	5171	4317	2857	2392	1787	1411	969	769
110	4431	3723	2490	2090	1565	1236	846	669
120	3820	3230	2182	1836	1378	1088	742	584

e. Case5: 4⁰ Curve

Speed in kmph	Level Track	Slope 1/500	Slope 1/200	Slope 1/150	Slope 1/100	Slope 1/75	Slope 1/50	Slope 1/40
START	6207	5446	3962	3432	2694	2206	1599	1312
10	6207	5446	3962	3432	2694	2206	1599	1312
15	6207	5446	3962	3432	2694	2206	1599	1312
20	6207	5446	3962	3432	2694	2206	1599	1312
30	6207	5446	3962	3432	2694	2206	1599	1312
40	6207	5446	3962	3432	2694	2206	1599	1312
50	6207	5446	3962	3432	2694	2206	1599	1312
60	6207	5446	3962	3432	2694	2206	1599	1312
70	6207	5446	3962	3432	2694	2165	1508	1214
80	6207	5446	3682	3087	2316	1838	1277	1024
90	5621	4718	3155	2652	1994	1582	1096	876
100	4793	4046	2731	2301	1733	1376	950	756
110	4119	3497	2382	2011	1518	1205	829	657
120	3561	3040	2089	1768	1337	1061	727	574

f. Case6: 5⁰ Curve

Speed in kmph	Level Track	Slope 1/500	Slope 1/200	Slope 1/150	Slope 1/100	Slope 1/75	Slope 1/50	Slope 1/40
START	5879	5189	3821	3324	2625	2158	1572	1293
10	5879	5189	3821	3324	2625	2158	1572	1293
15	5879	5189	3821	3324	2625	2158	1572	1293
20	5879	5189	3821	3324	2625	2158	1572	1293
30	5879	5189	3821	3324	2625	2158	1572	1293
40	5879	5189	3821	3324	2625	2158	1572	1293
50	5879	5189	3821	3324	2625	2158	1572	1293
60	5879	5189	3821	3324	2625	2158	1572	1293
70	5879	5189	3821	3324	2625	2112	1480	1194
80	5879	5189	3521	2971	2248	1793	1253	1008
90	5223	4430	3019	2553	1934	1543	1075	862
100	4465	3807	2615	2216	1682	1341	932	744
110	3847	3296	2282	1938	1473	1175	813	646
120	3334	2870	2003	1704	1298	1034	713	564

9 ENTRY-EXIT SPEEDS

Below are the tables indicating attacking speeds for a different load cases, gradients and lengths of track sections.

The track considered is dry track with an adhesion of 40%. The results are indicative in nature and actual values depends on the available adhesion between rail and wheel and actual load on the locomotive.

9.1 Load Case 1 (7600 T)

a. 1:100 compensated:

Gradient: 1:100 Compensated		
Attacking speed(kmph)	Max length which can be negotiated (km)	Exit speed(kmph)
20	0.5	17
30	1.5	16
40	2	16
50	3	16
60	4	16
70	5	16

b. 1:150 compensated:

Gradient: 1:150 Compensated		
Attacking speed(kmph)	Max length which can be negotiated (km)	Exit speed(kmph)
20	4	20
30	8	30
40	16	40
50	29	41
60	29	41
70	29	41

c. 1:200 compensated:

Gradient: 1:200 Compensated		
Attacking speed(kmph)	Max length which can be negotiated (km)	Exit speed(kmph)
20	4	20
30	8	30
40	16	40
50	16	50
60	16	60
70	29	64

9.2 Load case 2 (7040 T)

a. 1:100 compensated:

Gradient: 1:100 Compensated		
Attacking speed(kmph)	Max length which can be negotiated (km)	Exit speed(kmph)
20	0.5	16
30	1.5	17
40	3	15
50	4	16
60	5	17
70	6.5	15

b. 1:150 compensated:

Gradient: 1:150 Compensated		
Attacking speed(kmph)	Max length which can be negotiated (km)	Exit speed(kmph)
20	4	20
30	8	30
40	16	40
50	16	50
60	29	50
70	29	50

c. 1:200 compensated:

Gradient: 1:200 Compensated		
Attacking speed(kmph)	Max length which can be negotiated (km)	Exit speed(kmph)
20	4	20
30	8	30
40	16	40
50	16	50
60	16	60
70	29	68

9.3 Load case 3 (6600 T)

a. 1:100 compensated:

Gradient: 1:100 Compensated		
Attacking speed(kmph)	Max length which can be negotiated (km)	Exit speed(kmph)
20	1.5	15
30	3	15
40	4.5	15
50	6	15
60	7.5	15
70	8.5	15

b. 1:150 compensated:

Gradient: 1:150 Compensated		
Attacking speed(kmph)	Max length which can be negotiated (km)	Exit speed(kmph)
20	4	20
30	8	30
40	16	40
50	16	50
60	29	58
70	29	58

c. 1:200 compensated:

Gradient: 1:200 Compensated		
Attacking speed(kmph)	Max length which can be negotiated (km)	Exit speed(kmph)
20	4	20
30	8	30
40	16	40
50	16	50
60	16	60
70	16	70

9.4 Load case 4 (6000 T)

a. 1:100 compensated:

Gradient: 1:100 Compensated		
Attacking speed(kmph)	Max length which can be negotiated (km)	Exit speed(kmph)
20	4	20
30	29	22
40	29	22
50	29	22
60	29	22
70	29	22

b. 1:150 compensated:

Gradient: 1:150 Compensated		
Attacking speed(kmph)	Max length which can be negotiated (km)	Exit speed(kmph)
20	4	20
30	8	30
40	16	40
50	16	50
60	16	60
70	29	64

c. 1:200 compensated:

Gradient: 1:200 Compensated		
Attacking speed(kmph)	Max length which can be negotiated (km)	Exit speed(kmph)
20	4	20
30	8	30
40	16	40
50	16	50
60	16	60
70	16	70